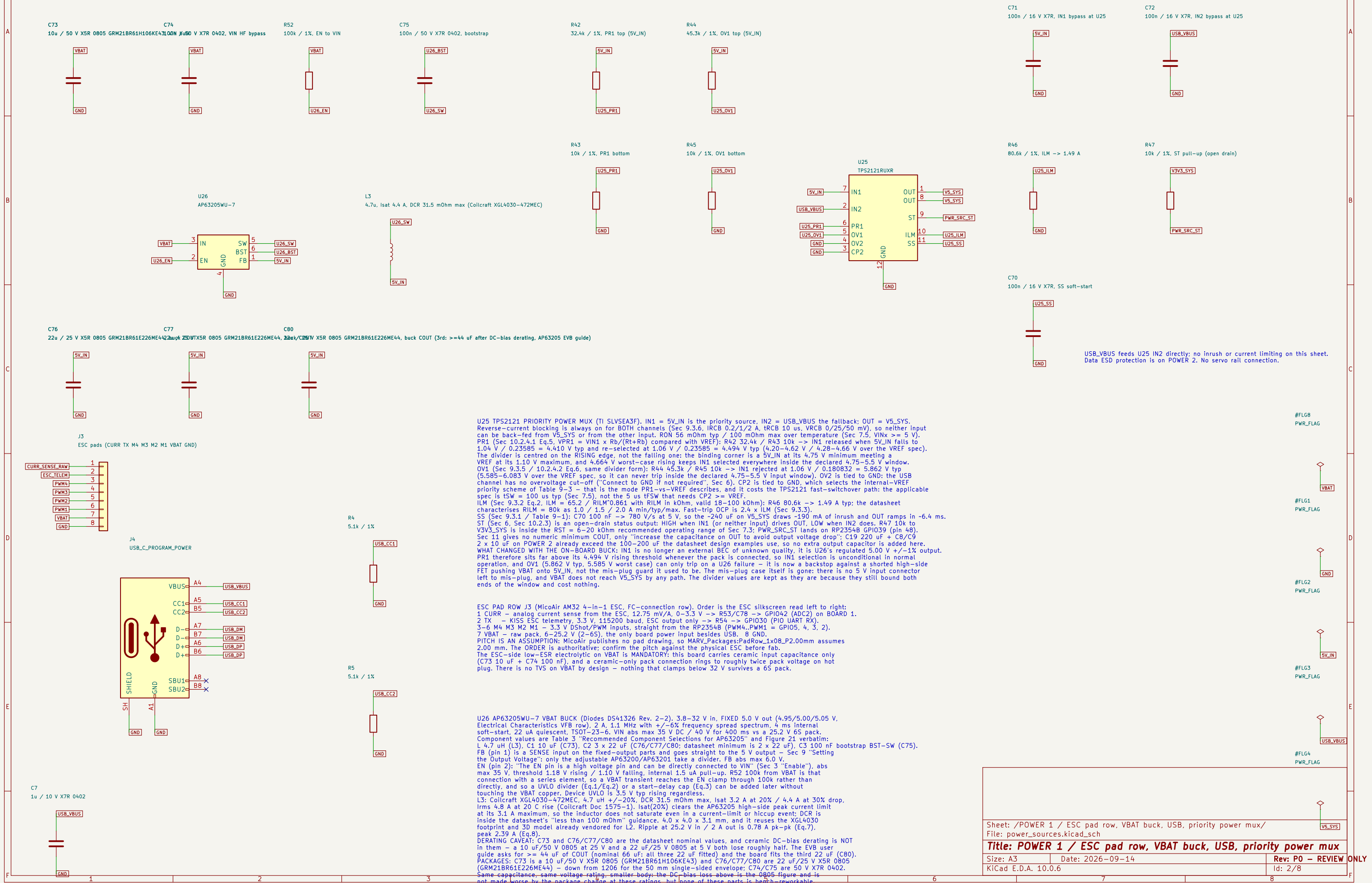


	1	2	3	4	5	6	7	8	
A	Integrated draft: power chain (sheets 2–4) and board stage (sheets 5–8). See DESIGN_SPEC.md. Inputs: VBAT 6–25.2 V (2–6S) from the MicoAir AM32 ESC pad row J3, and USB VBUS. Rails: 5V_IN from the U26 AP63205 buck, V5_SYS from the U25 TPS2121 mux, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. Servo 5 V is 5V_IN, upstream of the mux. No PCB placement or routing has been performed.								A
B	POWER 1 / ESC pad row, VBAT buck, USB, priority power mux File: power_sources.kicad_sch				BOARD 2 / IMU, barometer, high-g File: sensors.kicad_sch				B
C	POWER 2 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch				BOARD 3 / log flash and microSD File: storage.kicad_sch				C
D	POWER 3 / IO array File: power_periph.kicad_sch				BOARD 4 / servo outputs PWM5–8 File: actuators.kicad_sch				D
E									E
F	BOARD 1 / RP2354B File: mcu.kicad_sch								F
	1	2	3	4	5	6	7	8	

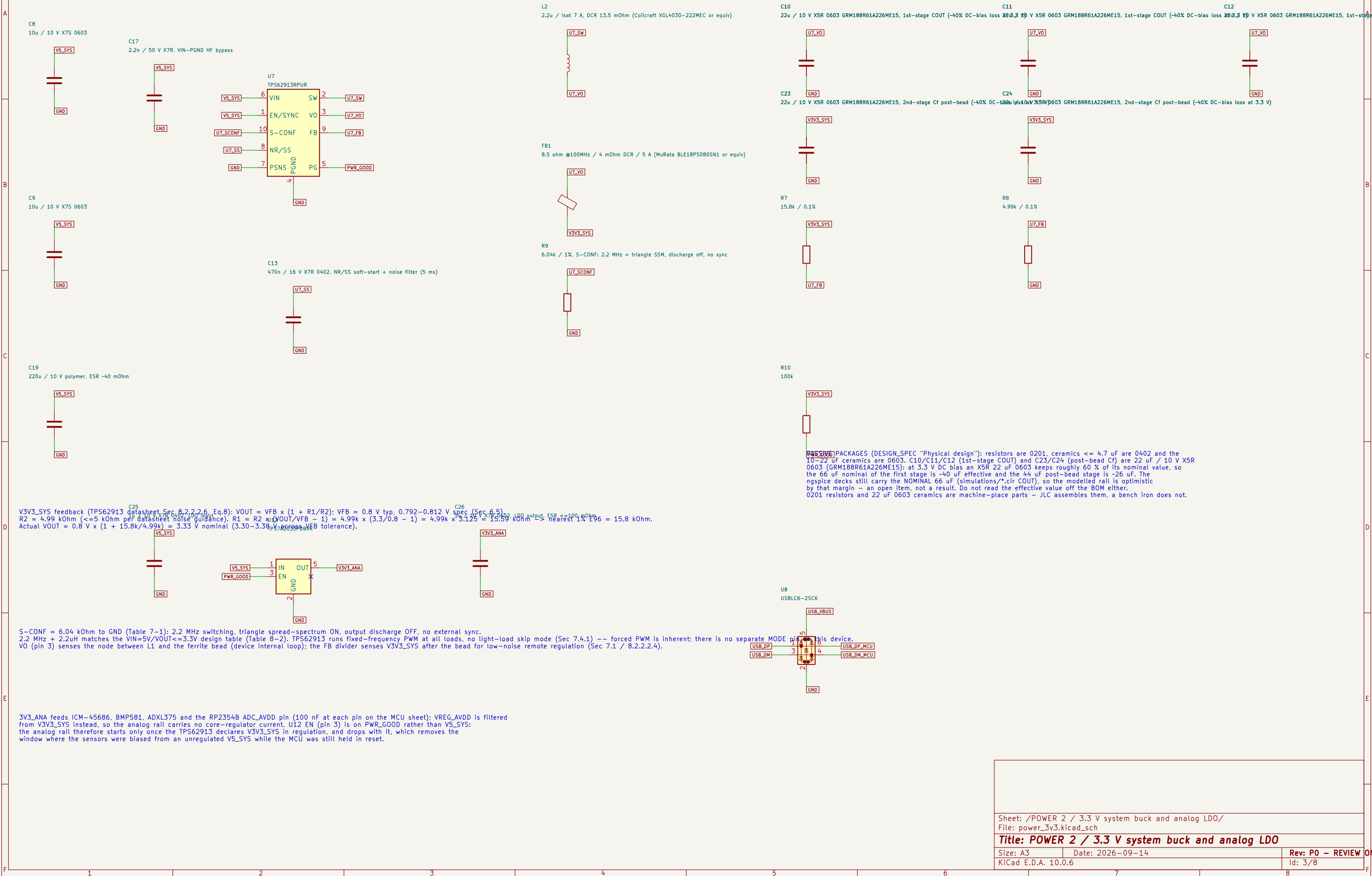
Sheet: / File: MARV-V2.kicad_sch		
Title: MARV integrated power – preliminary		
Size: A3	Date: 2026-09-14	Rev: P0 – NOT FOR
KiCad E.D.A. 10.0.6	Id: 1/8	

Sheet: / File: MARV-V2.kicad_sch		
Title: MARV integrated power – preliminary		
Size: A3	Date: 2026-09-14	Rev: P0 – NOT FOR FAB
KiCad E.D.A. 10.0.6	Id: 1/8	

BOARD INPUTS: VBAT 6–25.2 V (2–6S) from the ESC pad row J3, and USB VBUS ~5 V (J4). U26 (AP63205WU) makes 5V_IN from VBAT; 5V_IN is the priority input of the U25 power mux and also the servo 5 V bus.



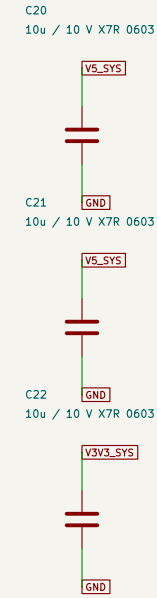
AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.



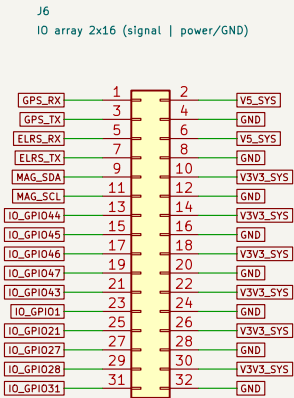
J6 IO ARRAY — ONE 2x16 THROUGH-HOLE BLOCK ON THE LEFT EDGE, 2.54 mm, madflight style. It replaces the thrge 1x04 solder-pad ports (the old J6 GPS / J7 ELRS / J9 MAG rows) and the old J16 2x6 spare-IO block: every signal that leaves this board except the ESC row (J3) and the SWD pads (J10) now leaves on this one connector.

COLUMN RULE: ODD pins are the OUTER column (signal, board edge side), EVEN pins the INNER column (power or ground), so every signal sits next to the rail its module wants and a 2-pin, a 3-pin or a full 2x16 shell all plug in the same way.

DS-009 NAMING IS KEPT: a port label is named for the PERIPHERAL pin it belongs to, so GPS_RX / ELRS_RX are module inputs driven by the RP2354B UART TX pins (rows 1 and 3, silk T0 / T1) and GPS_TX / ELRS_TX are module outputs that land on the MCU UART RX pins (rows 2 and 4, silk R0 / R1). The Pixhawk DS-009 ORDER of the old pad rows is gone with the pad rows: what survives is the naming convention and the rail assignment (GPS and ELRS on 5 V, magnetometer on 3.3 V).



C20 / C21: V5_SYS local bulk at the array (rows 1–4, the GPS and ELRS supply pins).
C22: V3V3_SYS local bulk at the array (rows 5–16).

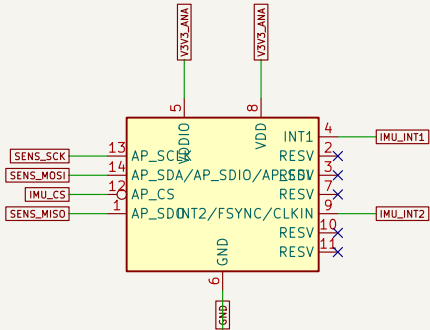


J6 ROW MAP (row n = pins 2n-1 odd / 2n even; silk label per row on the outer column)		
1	T0 GPS_RX	V5_SYS UART0 TX -> GPS RX (GPIO12)
2	R0 GPS_TX	GND UART0 RX <- GPS TX (GPIO13)
3	T1 ELRS_RX	V5_SYS UART1 TX -> ELRS RX (GPIO8)
4	R1 ELRS_TX	GND UART1 RX <- ELRS TX (GPIO9)
5	SDA MAG_SDA	V3V3_SYS I2C1 SDA (GPIO6), 4.7k pull-up on the MCU sheet
6	SCL MAG_SCL	GND I2C1 SCL (GPIO7), 4.7k pull-up on the MCU sheet
7	A44 IO_GPIO44	V3V3_SYS GPIO44 = ADC4
8	A45 IO_GPIO45	GND GPIO45 = ADC5
9	A46 IO_GPIO46	V3V3_SYS GPIO46 = ADC6
10	A47 IO_GPIO47	GND GPIO47 = ADC7
11	A43 IO_GPIO43	V3V3_SYS GPIO43 = ADC3
12	I01 IO_GPIO1	GND GPIO1
13	I021 IO_GPIO21	V3V3_SYS GPIO21
14	I027 IO_GPIO27	GND GPIO27
15	I028 IO_GPIO28	V3V3_SYS GPIO28 (the deleted buzzer driver used to own it)
16	I031 IO_GPIO31	GND GPIO31

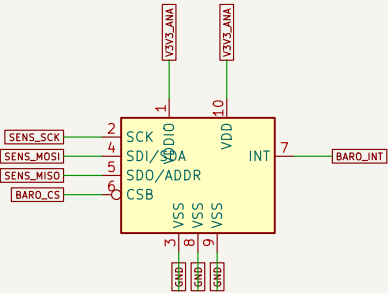
RAILS ON THE ARRAY: rows 1–4 (GPS, ELRS) take V5_SYS on their even pin — GNSS and ELRS modules are 5 V-powered with their own on-module regulators, and their C20 / C21 10 uF local bulk sits on V5_SYS with them; their SIGNAL pins stay 3.3 V CMOS straight from the RP2354B, which is what both module families expect. Rows 5–16 take V3V3_SYS: the magnetometer bus is 3.3 V and its pull-ups (R25/R26 on the MCU sheet) bias to V3V3_SYS, so a 5 V pin there would fight them, and the ten spare GPIOs are 3.3 V CMOS. Every even pin on rows 2, 4, 6, 8, 10, 12, 14, 16 is GND, so each signal has a return within one row of it. NO 5 V ANYWHERE BELOW ROW 4: a 3.3 V-only module cannot be fed 5 V by plugging it one row out of place, because rows 5–16 carry 3.3 V or ground and nothing else.

CURRENT: rows 1–4 are outside the V3V3_SYS budget and count against the U26 buck through the mux; rows 5–16 count in the V3V3_SYS budget (DESIGN_SPEC "Rails and load budget"). No fuse and no per-row current limit: the array is a system-integration connector, not a protected port.

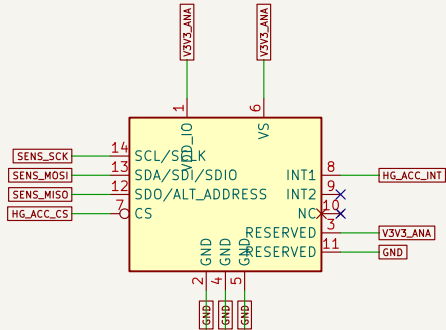
All three sensors run 4-wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TP57A20 LDO). Every supply pin on this sheet gets 100 nF + 1 uF; BMP581 and ADXL375 because their own datasheets call for the pair, the ICM-45686 as a deliberate addition – its datasheet BOM (DS-000577 Rev 1.0 Table 11) lists 0.1 uF only.



U21
ICM-45686 6-axis IMU



U22
BMP581 barometer

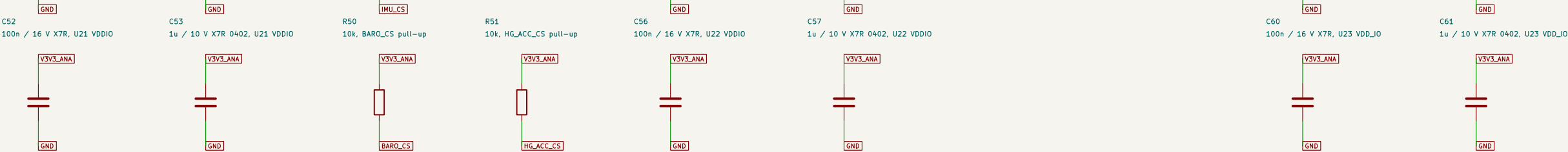


U23
ADXL375 high-g accelerometer

TP1 TP SENS_SCK TP2 TP SENS_MOSI TP3 TP SENS_MISO TP4 TP IMU_CS TP5 TP BARO_CS TP6 TP HG_ACC_CS TP7 TP IMU_INT1 TP8 TP IMU_INT2 TP9 TP BARO_INT TP10 TP HG_ACC_INT

C50 100n / 16 V X7R, U21 VDD C51 1u / 10 V X7R 0402, U21 VDD R48 10k, IMU_CS pull-up C54 100n / 16 V X7R, U22 VDD BARO_CS 55 1u / 10 V X7R 0402, U22 VDD HG_ACC_CS 100n / 16 V X7R, U23 VS IMU_INT2 1u / 10 V X7R 0402, U23 VS BARO_INT 1u / 10 V X7R 0402, U23 VS HG_ACC_INT

TEST POINTS TP1-TP10 (TestPoint:TestPoint_Pad_1.0x1.6mm, 1 x 1 mm bare F.Cu pads, 50 model by design – tools/audit footprints by --no-3d-ok covers TestPoint, along with the pad rows and mounting holes). TP1 SENS_SCK | TP2 SENS_MOSI | TP3 SENS_MISO | TP4 IMU_CS | TP5 BARO_CS | TP6 HG_ACC_CS | TP7 IMU_INT1 | TP8 IMU_INT2 | TP9 BARO_INT | TP10 HG_ACC_INT. Placement: Inside the sensor cluster, within 6 mm of U21/U22/U23 (tools/setup_pcb.py checks it). They add ten stubs to the SPI bus, which is why they are 1 mm pads next to the parts rather than a header somewhere else: at 10 MHz SCK a short stub is a capacitive load, a long one is a transmission line. No series resistors and no ground pad of their own – probe ground comes off a mounting hole or the J6 array.



U21 ICM-45686 (TDK InvenSense DS-000577 Rev 1.0, Table 10 "Signal Descriptions" / Figure 4 pin-out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = IMU_CS, AP_SCLK (13) = SENS_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = SENS_MOSI, AP_SDO/AP_ADO (1) = SENS_MISO, INT1 (4) = IMU_INT1, INT2/FSYNC/CLKIN (9) = IMU_INT2 – typed bidirectional because FSNC/CLKIN are host-driven inputs while INT2 is a chip output; only the INT2 function is used here. VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single-Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per-pin notes in Table 10); they are only needed for the AUX1 or I2C-master dual-interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 "Bill of Materials" (C1/C2 in the datasheet); C51 (VDD) and C53 (VDDIO) add 1 uF on top of them. The datasheet BOM lists 0.1 uF only; this board keeps 100 nF + 1 uF per supply pin for consistency with U22/U23 and common flight-controller practice.

U22 BMP581: CSB = BARO_CS, SDI = SENS_MOSI, SDO = SENS_MISO, SCK = SENS_SCK, INT = BARO_INT, VDDIO and VDD on V3V3_ANA, all three VSS pins to GND.
U23 ADXL375: CS = HG_ACC_CS, SDA/SDI = SENS_MOSI, SDO = SENS_MISO, SCL/SCLK = SENS_SCK, INT1 = HG_ACC_INT. Per the pin function table, pin 3 (RESERVED) goes to VS and pin 11 (RESERVED) to GND; pin 10 (NC) and the unused INT2 are left open. VS and VDD I/O are both on V3V3_ANA.

R48, R50, R51: 10k pull-ups to V3V3_ANA on the three SPI chip selects (IMU_CS, BARO_CS, HG_ACC_CS). R49 (the former second IMU chip-select pull-up) is removed along with that net – the ICM-45686 has a single chip select. Same rationale as R35 on FLASH_CS1 (BOARD 3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull-up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

Sheet: /BOARD 2 / IMU, barometer, high-g/
File: sensors.kicad_sch

Title: BOARD 2 / IMU, barometer, high-g

Size: A3 Date: 2026-09-14

KiCad E.D.A. 10.0.6

Rev: P0 – REVIEW ONLY

Id: 6/8

Sheet: /BOARD 3 / log flash and microSD/			ONLY
File: storage.kicad_sch			
Title: BOARD 3 / log flash and microSD			
Size: A3	Date: 2026-09-14	Rev: P0 - REVIEW	
KiCad E.D.A. 10.0.6		Id: 7/8	

F

Diagram showing four 5V_IN pins connected to a 4-pin header.

A diagram showing four pins of a 4-pin header, numbered 1 to 4. Each pin is connected to a GND pin of a component, represented by a box labeled 'GND'.

Three stacked Conn_01x04 symbols (J12 signal, J13 5 V, J14 GND) with three 1x04 THT footprints: no 3-row generic connector symbol or 2.54 mm 3x04 footprint ships with this KiCad install. Place them on 2.54 mm centres at the bottom-left of the board to form a 3x4 servo block in standard servo lead order (SIG / 5 V / GND); column n is J12 pin n / J13 pin n / J14 pin n. Through-hole 2.54 mm is deliberate here: DESIGN_SPEC "Physical design" puts solder pads on every bus and keeps THT pins for leftover IO only.

```
ON THE ESC PAD ROW (J3, POWER 1)
PWM1 = GPIO2
PWM2 = GPIO3
PWM3 = GPIO4
PWM4 = GPIO5
```

Id: 8/8